Nguyen Van Cuong

AI Engineer | Former Web Developer (3+ years experience)

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OBJECTIVE

- AI Engineer with strong passion for technology and 3 years of experience in system development (Web Development). Currently building hands-on projects in Deep Learning, Computer Vision, NLP, and LLM applications. Eager to apply artificial intelligence to solve real-world problems and create value for the community. Seeking opportunities in a high-tech environment that encourages collaboration and problem-solving to help organizations achieve their AI strategic goals.

SKILLS

Programming	Python, JavaScript, TypeScript
Machine Learning / Deep Learning	ML Algorithms, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM
LLM & RAG	HuggingFace, LangChain, LlamaIndex, RAG, AI Agents
Computer Vision	OpenCV, YOLO
Data Processing	Pandas, NumPy, Matplotlib
Application & Deployment	React, Next.js, Strapi

WORK EXPERIENCE

Derasoft Co., Ltd 2022 - 2025

Web Developer

- Collaborated with clients to understand requirements, solve technical issues, and deliver effective web solutions.
- Developed and maintained websites using ReactJS, NextJS, and Strapi CMS, including Headless CMS integration and CI/CD deployment.
- Optimized website performance, SEO, and UX/UI while supporting data-driven improvements with cross-functional teams.
- Designed and integrated REST APIs and backend services to support scalable application layers and data workflows.
- Applied automation practices and API integrations that support building AI-powered applications.

PROJECTS

Vietnamese Emotion Recognition System | Python, Scikit-learn, SVM, NLP 2025 - 2026

Personal Project

- Engineered a high-performance sentiment analysis system to classify Vietnamese text into 7 emotional states (Anger, Disgust, Enjoyment, Fear, Other, Sadness, Surprise) with **92.7% accuracy**.
- **Advanced NLP Pipeline:** Architected a comprehensive preprocessing workflow tailored for Vietnamese, including abbreviation normalization, profanity filtering, and emoji-to-emotion token mapping. Utilized underthesea for precise Vietnamese word segmentation.
- **Data Optimization:** Addressed class imbalance issues by implementing **Oversampling** techniques, ensuring the model's robustness across all emotional categories.

- **Feature Engineering & Modeling:** Leveraged **TF-IDF Vectorization** with N-gram (1,2) features to capture local context. Trained a **Linear SVM** model, optimized for high-dimensional sparse text data.
- **Evaluation:** Achieved a superior **F1-macro score of 0.927**, demonstrating the system's ability to handle complex linguistic nuances and "noisy" social media text.
- Link: *https://colab.research.google.com/drive/1ACDiTGC1zl3V1wWctdJCJy265yb68Zpc?usp=sharing*

Airbnb Price Prediction | Python, Scikit-learn, CatBoost 2025 - 2026

Personal Project

- Built an **Airbnb price prediction model** using listing features such as property type, location, amenities, and reviews.
- **Data Processing:** Cleaned data, handled missing values, encoded categorical variables, and converted amenities into numerical features.
- **Modeling:** Trained multiple regression models (**Linear Regression, Random Forest, Gradient Boosting, CatBoost**) and selected **CatBoost** as the best model.
- **Result:** Achieved **$R^2 \approx 0.70$** and deployed the model using **Scikit-learn pipeline and Flask API**.
- Link: *https://github.com/ncuong0704/About-me/tree/main/nlp/End-to-End-Airbnb-Price-Prediction*

Real-time Poker Hand Recognition System | YOLOv8, OpenCV, PyTorch 2025 - 2026

Personal Project

- Built a real-time computer vision system to detect **52 playing card classes** from webcam input.-
- Fine-tuned **yolov8m.pt** on the **Roboflow Playing Cards dataset**, achieving **mAP@50: 0.96** and **mAP@50-95: 0.82**.
- Implemented Poker hand evaluation logic to classify hands such as **Straight, Flush, Full House,** and **Royal Flush**.
- Used **OpenCV** for live stream processing and real-time visualization.
- Link: *https://github.com/ncuong0704/About-me/tree/main/computer_vision/playing_cards_detection*

Agentic RAG Agent (Local Execution) | Ollama, Llama 3.2, LanceDB, Streamlit 2025 - 2026

Personal Project

- Engineered an agentic Retrieval-Augmented Generation (RAG) system to enable intelligent querying and interaction with PDF-based knowledge bases.
- **Local LLM Integration:** Leveraged **Llama 3.2** and **EmbeddingGemma** via **Ollama** to ensure data privacy and efficient local inference.
- **Advanced Vector Search:** Implemented **LanceDB** for high-performance vector storage and semantic search, enabling dynamic document ingestion from PDF URLs.
- **Agentic Workflow:** Developed a tool-calling logic that allows the model to autonomously decide when to retrieve information or synthesize answers.
- **Real-time Interface:** Designed a **Streamlit** dashboard featuring streaming responses and "thought-process" visibility for enhanced user interaction.
- Link: *https://github.com/ncuong0704/About-me/tree/main/rag/agentic_rag_embedding_gemma*

EDUCATION

Ho Chi Minh City University of Industry and Trade (HUIT) 2013 - 2016

Bachelor of Business Administration

- Focused on: *Statistics, Excel-based Data Analysis, Market Research, and Business Decision-Making.*

CERTIFICATIONS

- Ranked Top 1 in Applied Machine Intelligence Specialist cohort – Cybersoft (2025)
- Ranked Top 5 in Fundamentals of Data Analyst and AI Automation cohort – Cybersof (2025)
- Data Science/Machine Learning/Python Advanced – Viet Nguyen AI (2025)
- Deep Learning for Computer Vision (Basic & Advanced) – Viet Nguyen AI (2025)
- Certificate of Proficiency in Information System Management (CPISM) - *Aptech (2022)*

COURSES IN PROGRESS

- Advanced AI Engineering with NLP, LLM, RAG, RAGFlow, LangChain, Generative AI, and AI Agents - *Cypersoft*
- ML Ops/Cloud – AI Deployment - *Cypersoft*

INTERESTS

- Playing sports (Football and badminton)

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